Matrices: Engineering Mathematics Study Notes

Iterative Methods

High-Level Overview

Unlike direct methods (such as Gauss-Elimination or Gauss-Jordan) which perform a fixed number of calculations to arrive at an exact solution, **iterative methods** start from an initial approximation of the true solution and generate a sequence of increasingly accurate approximations through a repeated computation cycle. The cycle continues until the required accuracy or convergence is reached.

Stopping Criteria for Iterations

We stop the iteration procedure when the magnitude of the difference between two successive iterates for all variables falls below a predefined error tolerance (ϵ):

$$\left|x_i^{(k+1)} - x_i^{(k)}\right| \le \epsilon \quad \text{for all } i$$

- **Two decimal places of accuracy:** Iterate until $\left|x_i^{(k+1)} - x_i^{(k)}\right| \le 0.005$ for all i.
- **Three decimal places of accuracy:** Iterate until $\left|x_i^{(k+1)} - x_i^{(k)}\right| \le 0.0005$ for all i.

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1. Jacobi Iteration Method

Core Principles

Also known as the **Gauss-Jacobi method** or the **method of simultaneous displacement**, this technique relies on rearranging the equations to isolate each variable on the diagonal.

- **Pivoting Condition:** We assume the diagonal elements are non-zero ($a_{ii} \neq 0$). If not, equations must be rearranged.
- **Condition for Convergence:** A sufficient condition for convergence is that the system is **diagonally dominant**, meaning the magnitude of the diagonal element in each row is greater than or equal to the sum of the magnitudes of all other elements in that row:

$$|a_{ii}| \ge \sum_{j=1, j\neq i}^{n} |a_{ij}|$$

- **Initial Approximation:** If no suitable approximation is available, we typically choose $x_i^{(0)} = 0$ for all variables.
- **Iteration Formula:** For a system rewritten to solve for individual variables, the $(k+1)^{\text{th}}$ iteration uses values exclusively from the previous k^{th} iteration:

$$x_1^{(k+1)} = \frac{1}{a_{11}}\left[b_1 - a_{12}x_2^{(k)} - a_{13}x_3^{(k)} - \cdots - a_{1n}x_n^{(k)}\right]$$

$$x_2^{(k+1)} = \frac{1}{a_{22}}\left[b_2 - a_{21}x_1^{(k)} - a_{23}x_3^{(k)} - \cdots - a_{2n}x_n^{(k)}\right]$$

$$\vdots$$

$$x_n^{(k+1)} = \frac{1}{a_{nn}}\left[b_n - a_{n1}x_1^{(k)} - a_{n2}x_2^{(k)} - \cdots - a_{n(n-1)}x_{n-1}^{(k)}\right]$$

Worked Examples

Example 1: Solving a 3×3 System via Jacobi Method

Solve the following system correct to two decimal places:

$$10x + y - z = 11.19$$

$$x + 10y + z = 28.08$$

$$-x + y + 10z = 35.61$$

Step 1: Verify Diagonal Dominance and Rearrange Equations
The system is diagonally dominant since $|10| \geq |1 + (-1)|$, $|10| \geq |1 + 1|$, and $|10| \geq |-1 + 1|$. We express the system to isolate $x, y,$ and z:

$$x = \frac{1}{10}(11.19 - y + z)$$

$$y = \frac{1}{10}(28.08 - x - z)$$

$$z = \frac{1}{10}(35.61 + x - y)$$

Step 2: Set Initial Approximations
Let $x^{(0)} = 0,\ y^{(0)} = 0,\ z^{(0)} = 0$.

Step 3: Perform Iterations

- **First Iteration:**

$$x^{(1)} = \frac{1}{10}(11.19 - 0 + 0) = 1.119$$

$$y^{(1)} = \frac{1}{10}(28.08 - 0 - 0) = 2.808$$

$$z^{(1)} = \frac{1}{10}(35.61 + 0 - 0) = 3.561$$

- **Second Iteration:**

$$x^{(2)} = \frac{1}{10}(11.19 - 2.808 + 3.561) = 1.1942$$

$$y^{(2)} = \frac{1}{10}(28.08 - 1.119 - 3.561) = 2.34$$

$$z^{(2)} = \frac{1}{10}(35.61 + 1.119 - 2.808) = 3.3921$$

- **Third Iteration:**

$$x^{(3)} = \frac{1}{10}(11.19 - 2.34 + 3.3921) = 1.2242$$

$$y^{(3)} = \frac{1}{10}(28.08 - 1.1942 - 3.3921) = 2.3493$$

$$z^{(3)} = \frac{1}{10}(35.61 + 1.1942 - 2.34) = 3.4464$$

- **Fourth Iteration:**

$$x^{(4)} = \frac{1}{10}(11.19 - 2.3493 + 3.4464) = 1.2287$$

$$y^{(4)} = \frac{1}{10}(28.08 - 1.2242 - 3.4464) = 2.3409$$

$$z^{(4)} = \frac{1}{10}(35.61 + 1.2242 - 2.3493) = 3.4484$$

Final Answer:

Rounding to two decimal places:

$$x = 1.22, \quad y = 2.34, \quad z = 3.44$$

Example 2: Performing 4 Iterations

Carry out 4 iterations of the Jacobi method for the system:

$$20x + y - 2z = 17$$

$$3x + 20y - z = -18$$

$$2x - 3y + 20z = 25$$

Step 1: Isolate Variables

$$x = \frac{1}{20}(17 - y + 2z)$$

$$y = \frac{1}{20}(-18 - 3x + z)$$

$$z = \frac{1}{20}(25 - 2x + 3y)$$

Step 2: Tabulate Iteration Results (Starting with $x^{(0)} = y^{(0)} = z^{(0)} = 0$)

Iteration (n)	x	y	z
1	0.85	-0.91	1.25
2	1.02	-0.965	1.03
3	1.00125	-1.0015	1.00325
4	1.0004	-1.00025	0.99965

Final Answer:

$$x = 1, \quad y = -1, \quad z = 1$$